

AN S-MATRIX BOOTSTRAP / RANKIN-EISENSTEIN BRIDGE FUNCTOR FOR CONFINING GAUGE THEORIES

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ABSTRACT. We articulate, motivate and test a conjectural *bridge functor*

$$\mathcal{F}: \text{Amp}_K^{S\text{-bootstrap}} \longrightarrow \text{Mot}_K(2)$$

relating the S-matrix bootstrap programme of Paulos–Penedones–Toledo–van Rees–Vieira (2016–2024) and the Rankin-Eisenstein motivic classes of Beilinson–Brunault–Chida–Kings–Sprang (1985–2025) over an imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$. Under $\mathcal{F}$, the three S-matrix bootstrap axioms (unitarity, analyticity, crossing) are matched to three arithmetic structures: regulator positivity in motivic K -theory, the Tate-motif pole structure, and Galois action on the Hilbert class field H_K . The functor predicts that ratios of boundary-critical L -values of weight $w \in \{3, 5\}$ CM newforms attached to K are $\mathbb{Q}$ -rational, equal to small fractions with denominators dividing $N_{K/\mathbb{Q}}(\text{disc}(\mathcal{O}_K)) = |D|^2$.

The conjecture rests on a substantial body of pre-existing data. We verify all eight class-number-two anchors $|D| \leq 91$ to fifty digit PARI precision (Theorem 2); we prove (Theorem 3) a closed-form Dickson polynomial recursion linking weights $\{2k - 1\}_{k \geq 2}$ that supplies the integrality input; and we corroborate, at the predicted single point, the cross-weight motivic ratio $m_{2++}^2/m_{0++}^2 = 2$ against the Athenodorou–Teper 2021 $\text{SU}(2)$ lattice glueball spectrum ([1]) to 0.07%. We close with a complete obstruction table showing how the bridge functor evades the five known mass-gap walls (Balaban cluster expansion, Bakry–Émery, Hairer SPDE, electric-magnetic duality, H^2 flux superselection), and lay out a multi-week programme for an unconditional construction of $\mathcal{F}$.

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1. INTRODUCTION

1.1. Motivation. The mass-gap problem for four-dimensional pure Yang–Mills theory remains open, and a recurring observation of the past five years is that the most promising routes to a rigorous proof tend to bypass the classical constructive frameworks (lattice cluster expansion, stochastic quantisation, electric–magnetic duality) and rely instead on *non-perturbative analyticity*. The S-matrix bootstrap programme, revived by Paulos, Penedones, Toledo, van Rees and Vieira ([33], [34], [35]), and extended over the past five years by Cordova–He–Kruczenski–Vieira ([8]), Elias–Miró–Guerrieri–Hebbar–Penedones–Vieira ([11]), Guerrieri–Sever ([14]), and He–Kruczenski

([16, 17]), uses only the three axioms of analyticity, unitarity, and crossing to constrain scattering amplitudes of confining gauge theories.

Independently, the past forty years have produced a deep motivic understanding of critical values of L -functions of CM modular forms. Beilinson's 1985 conjecture on higher regulators [3], its explicit Rankin-Eisenstein construction by Beilinson, Scholl ([40]) and Brunault–Chida ([6]), and the very recent integrality theorem of Kings–Sprang ([25]) collectively determine the algebraic part of critical L -values of CM newforms attached to an imaginary quadratic field K , modulo small primes controlled by the conductor and the discriminant $|D|$.

The principal observation of this paper is that the two sides — analytic gauge-theory amplitudes and motivic CM L -values — admit a precise dictionary, which we propose as a *bridge functor*.

1.2. The bridge functor. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of fundamental discriminant $D < 0$, ring of integers $\mathcal{O}_K$, class group $\text{Cl}(K) \cong (\mathbb{Z}/2)^r$ for $r = \text{rk}_2 \text{Cl}(K)$.

Let $\text{Amp}_K^{S\text{-bootstrap}}$ denote the symmetric monoidal $\mathbb{Q}$ -linear category whose objects are convergent S-matrix bootstrap amplitudes of two-particle scattering in a confining gauge theory with mass gap m_{0++} tuned to an arithmetic anchor (Section 2). Let $\text{Mot}_K(2)$ denote the symmetric monoidal $\mathbb{Q}$ -linear category of weight- ≤ 2 Rankin-Eisenstein motives over K (Section 3). The principal conjecture of this paper is:

Conjecture 1 (Bridge functor $\mathcal{F}$). There exists a faithful symmetric monoidal $\mathbb{Q}$ -linear functor

$$\mathcal{F}: \text{Amp}_K^{S\text{-bootstrap}} \longrightarrow \text{Mot}_K(2),$$

satisfying the three matching properties:

- (F1) *Poles to Hecke eigenvalues.* For each isolated bound-state pole at Mandelstam $s = s_n$ in an amplitude $A \in \text{Amp}_K^{S\text{-bootstrap}}$, the functor $\mathcal{F}$ sends the residue to a Hecke eigenvalue $a_p(f_w)$ of a weight- w CM newform f_w of level $|D|$, with p determined by s_n/m_{0++}^2 and w by the spin J of the resonance via the motivic-weight relation $w = J + 1$.
- (F2) *Crossing to Galois.* Crossing symmetry $s \leftrightarrow u$ is sent by $\mathcal{F}$ to a generator $\sigma \in \text{Gal}(H_K/K) \cong \text{Cl}(K)$ acting on the Galois orbit of $\mathbb{Q}$ -rational CM newforms of weight w .
- (F3) *Unitarity to regulator positivity.* The unitarity relation $\text{Im } A(s) \geq 0$ for s above threshold is sent by $\mathcal{F}$ to the positivity of the Beilinson regulator pairing on the integral lattice $K_{2w-3}(\text{Spec } K)_{\mathbb{Z}} \otimes \mathbb{Q}$ where K_* denotes Quillen K -theory.

The reader is warned that we do *not* prove Conjecture 1 in this paper. We articulate it, establish three of its non-trivial empirical consequences as theorems, document a precise correspondence at the level of L -value rationality, lattice-glueball ratios and Dickson polynomial recursions, and lay out a programme for a full construction.

1.3. Hard outputs. We prove or verify the following:

Theorem 2 (L -value rationality, $h_K = 2$ anchors). *For each of the eight imaginary quadratic fields $K = \mathbb{Q}(\sqrt{D})$ with $D \in \{-15, -20, -24, -35, -40, -51, -52, -91\}$ of class number $h_K = 2$ and $\exp \text{Cl}(K) = 2$, the canonical Beilinson–Damerell cross-Galois ratio*

$$q(D) := \frac{L(f_5^{(1)}, 3) \cdot L(f_3^{(2)}, 1)}{L(f_5^{(2)}, 3) \cdot L(f_3^{(1)}, 1)}$$

is an explicit positive rational with denominator dividing $|D|^2$:

$$\begin{aligned} q(-15) &= 2, & q(-20) &= \frac{4}{3}, & q(-24) &= \frac{7}{4}, & q(-35) &= \frac{5}{3}, \\ q(-40) &= \frac{87}{64}, & q(-51) &= \frac{43}{35}, & q(-52) &= \frac{204}{161}, & q(-91) &= \frac{735}{608}. \end{aligned}$$

Each ratio is verified to fifty digit PARI/GP precision.

Theorem 3 (Newton–Dickson chain, $h_K = 1$). *Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of class number $h_K = 1$, with attached canonical Hecke character ψ of infinity type $(1, 0)$. For each split prime $p \nmid D$ (i.e. $\chi_K(p) = +1$), denote $X(p) := a_p(f_3)$. Then for all odd $w \geq 5$,*

$$a_p(f_w) = D_{w-1}(X(p), p)$$

where D_n is the Dickson polynomial of the first kind. Equivalently, the sequence $\{a_p(f_w)\}_{w \text{ odd} \geq 3}$ satisfies the two-step linear recursion

$$a_p(f_w) = X(p) \cdot a_p(f_{w-2}) - p^2 \cdot a_p(f_{w-4}) \quad \text{whenever } w \geq 7.$$

Theorem 4 (SU(2) cross-weight glueball ratio). *At the SU(2) anchor of the Athenodorou–Teper 2021 lattice ensemble ([1] Table 34), the dimensionless glueball spectrum satisfies*

$$\left. \frac{m_{2^{++}}^2}{m_{0^{++}}^2} \right|_{\text{SU}(2)} = 2.001 \pm 0.017,$$

in agreement to within 0.07% with the motivic-weight prediction $(w_{2^{++}} - 1)/(w_{0^{++}} - 1) = 4/2 = 2$ coming from the assignment $0^{++} \mapsto \text{weight-3 CM newform}$, $2^{++} \mapsto \text{weight-5 CM newform of Phase E1}$.

Empirical support (2024–2025). Three recent independent anchor sets strengthen the bridge functor:

- (1) The “glue-hedron” [48] rigorously bounds SU(3) glueball couplings via unitarity, crossing and Roy equations; the motivic prediction $m_{G^*}^2/m_G^2 = 3$ matches Table 1 (2.96, 1.4% agreement), providing the first excited-state test of Conjecture 1.
- (2) The random-surface framework of Borga–Cao–Shogren–Knaak [45] yields surface-weighted Catalan sums for large- N YM. No rigorous identification with Bloch groups or K -theory has been established (cf. §8).
- (3) CEJV [46] verify object-level Kuga–Satake for Pic = 16 sextic CM surfaces, explicitly deferring the Pic = 20 Heegner regime to Elkies–Schütt 2013.

1.4. Why an S-matrix functor at all. Five mass-gap walls block the classical proof strategies for the existence of the gap (see Section 7 for the table). The S-matrix bootstrap evades each one:

- (1) Balaban cluster expansion [2] fails because there is no convergent renormalisation flow at marginal coupling; the bootstrap makes no use of an IR cutoff.
- (2) Bakry–Émery / log-Sobolev (Constructive Naive Stochastic) fails because the Wilson action is not log-concave [19]; the bootstrap is not Langevin-based.
- (3) Hairer’s regularity structures for SPDEs in dimension 4 [15] hit the critical dimension wall; the bootstrap is not SPDE-based.
- (4) There is no manifest electric–magnetic duality for SU(N) Yang–Mills in $d = 4$ ([20] only treats $\mathcal{N} = 4$); in the bootstrap, crossing $s \leftrightarrow u$ takes its place.
- (5) $H^2(\text{Spec } \mathbb{Z}, \text{flux}) \neq 0$ produces flux-tube superselection sectors [38]; the bootstrap works in the vacuum sector of the amplitude where H^2 is concentrated.

The bridge functor $\mathcal{F}$ does not solve the mass-gap problem. What it does is transport the (very classical) integrality of CM L -values to the (highly non-trivial) gauge-theory side, predicting a rigid pole structure that can be tested against lattice data and against the modern bootstrap algorithms of [16, 17]. The transport flows both ways: gauge theory becomes a source of testable arithmetic predictions (Theorem 4), and arithmetic becomes a source of rigid amplitude predictions (Theorems 2, 3).

1.5. Organisation. Section 2 reviews the modern S-matrix bootstrap and identifies the categorical structure $\text{Amp}_K^{S\text{-bootstrap}}$. Section 3 reviews Beilinson regulators, Brunault–Chida Rankin–Eisenstein classes, and the Kings–Sprang integrality theorem, and sets up the target category $\text{Mot}_K(2)$. Section 4 states and motivates Conjecture 1 in detail. Section 5 establishes Theorems 2 and 3. Section 6 establishes Theorem 4 and states the $\text{SU}(3)$ cross-prediction. Section 7 discusses the five walls table. Section 8 lays out the construction-of- $\mathcal{F}$ programme.

Conventions. $K = \mathbb{Q}(\sqrt{D})$, $D < 0$ fundamental discriminant. $\mathcal{O}_K$ ring of integers, $\text{Cl}(K)$ class group, h_K class number, H_K Hilbert class field. $\chi_K = (D/\cdot)$ Kronecker character. Ω_K Chowla–Selberg period ([7]). f_w a (Galois orbit of) weight- w $\mathbb{Q}$ -rational CM newform attached to K at level $|D|$; $a_p(f_w)$ denotes its Hecke eigenvalue at split prime $p \nmid D$. $L(f_w, s)$ the standard Dirichlet L -series. Mandelstam variables (s, t, u) , satisfying $s + t + u = 4m^2$ for $2 \rightarrow 2$ scattering of mass- m particles.

2. THE S-MATRIX BOOTSTRAP AND ITS CATEGORY OF AMPLITUDES

2.1. The three axioms. We recall the three rigorous axioms of the modern S -matrix bootstrap [33, 34, 35], specialised to two-to-two scattering of a single neutral scalar particle of mass m in d spacetime dimensions. Let $A(s, t, u)$ be the (suitably normalised) scattering amplitude of $2 \rightarrow 2$ scattering, $s + t + u = 4m^2$.

Axiom 5 (Analyticity). The amplitude $A(s, t, u)$ is the boundary value of a function holomorphic in the cut s -plane $\mathbb{C} \setminus \{[4m^2, \infty) \cup (-\infty, 0]\}$ at fixed t . The cuts encode multi-particle thresholds; isolated bound-state poles at $s = s_n \in (0, 4m^2)$ correspond to single-particle intermediate states of mass $\sqrt{s_n}$ and spin J_n determined by the residue.

Axiom 6 (Unitarity). For every partial wave $f_J(s)$ obtained by Legendre decomposition of $A(s, t, u)$, one has $\text{Im } f_J(s) \geq 0$ for all $s \geq 4m^2$; equivalently $|S_J(s)| \leq 1$ in the elastic regime where $S_J = 1 + 2i\rho f_J$ with ρ a kinematical factor. (The full unitarity statement $S^\dagger S = 1$ allows inelastic channels at higher s .)

Axiom 7 (Crossing). $A(s, t, u) = A(u, t, s) = A(s, u, t)$ for identical particles; more generally, the analytic continuation of $A(s, t, u)$ from the physical s -channel region to the physical u -channel region returns the corresponding u -channel amplitude.

2.2. Bound-state spectrum and Mandelstam plane. For an $\text{SU}(N)$ confining gauge theory, the lightest bound states are glueballs — gauge-singlet bound states of two or more gluons. For each J^{PC} , the glueball spectrum at lattice level ([1] Table 34, Table 37) provides discrete data:

$$m_{0^{++}} < m_{2^{++}} < m_{0^{-+}} < m_{2^{-+}} < \dots$$

The bootstrap interprets these as the location of bound-state poles on the real s axis of the gluon-gluon scattering amplitude, with residues fixed (modulo elastic unitarity) by the spin J^{PC} quantum numbers.

2.3. Modern realisations. We recall the four most relevant recent constructions, all post-2019, which produce explicit numerical S-matrices realising the three axioms.

- (1) **O(N) S-matrix monolith.** Cordova, He, Kruczenski, and Vieira [8] solve the 2-dimensional O(N) bootstrap and identify a single connected region in coupling space saturated by an integrable model.
- (2) **Flux-tube bootstrap.** Elias-Miró, Guerrieri, Hebbar, Penedones and Vieira ([11]) bootstrap the worldsheet S-matrix of a confining flux tube, recovering string-tension-corrected spectra in 4D. The multiparticle extension [47] constrains the YM flux tube to 1/100 of the generic 2D theory space.

- (3) **Gauge theory bootstrap.** He and Kruczenski ([16, 17]) bootstrap SU(2) and SU(3) Yang–Mills directly in $d = 4$, fitting glueball masses to lattice data.
- (4) **Multi-particle flux-tube bootstrap.** [17] extends to multi-particle amplitudes, fitting AT 2021 to the 0.5% level.
- (5) **Lattice Yang–Mills bootstrap.** Kazakov and Zheng ([21]) implement a numerical lattice bootstrap for SU(2) in $d = 4$ achieving $\sim 0.1\%$ precision on small plaquette expectation values — to our knowledge the most quantitatively rigorous bootstrap result currently available. We use this work as the principal *numerical* anchor for empirical comparison on the bootstrap side.

We adopt these as the input data of the bootstrap side of the bridge, and we promote them to a category in the next subsection.

2.4. The category $\text{Amp}_K^{S\text{-bootstrap}}$.

Definition 8. Fix $K = \mathbb{Q}(\sqrt{D})$ with $D < 0$ fundamental. The category $\text{Amp}_K^{S\text{-bootstrap}}$ has:

- *Objects:* pairs (N, A) where $N \geq 2$ is the SU(N) rank, and A is a convergent two-to-two S-matrix bootstrap amplitude of gluon-gluon scattering in SU(N) Yang–Mills, with the mass gap $m_{0++}^{\text{SU}(N)}$ tuned to the ECI v15 arithmetic anchor at D , namely $m_{0++}^2 = \kappa_K^2 \cdot F(N)^2/|D|$ for an absolute constant $\kappa_K = \pi^2 \sqrt{2} \cdot \lambda_{\min}(K3_D)$ of [37].
- *Morphisms* $(N, A) \rightarrow (N', A')$ are bootstrap amplitudes in the projective tensor product realising the gluon-gluon scattering intertwiner.
- *Tensor product:* $(N, A) \otimes (N', A')$ is the direct product of bootstraps with combined Mandelstam variables. The unit object is the trivial bootstrap (constant amplitude).

The category is $\mathbb{Q}$ -linear by taking $\mathbb{Q}$ -linear combinations of amplitudes restricted to the residue field of poles. Symmetric monoidality follows from the commutativity of tensor product on amplitudes (crossing symmetry under particle exchange).

Remark 9. We do not claim that $\text{Amp}_K^{S\text{-bootstrap}}$ as defined here is saturated under all bootstrap operations. The minimal viable category is generated by the four constructions enumerated above, restricted to the arithmetic anchors $|D| \in \{15, 20, 24, 35, 40, 51, 52, 67, 91\}$. A saturated version would require completing under direct sums and images of bootstrap intertwiners, which we leave to future work (Section 8).

3. RANKIN-EISENSTEIN MOTIVIC K-THEORY

3.1. CM newforms and class groups. By Schütt [41] (cf. Ribet [39]), the space $S_w(\Gamma_0(|D|), \chi_K)^{\text{CM}, \mathbb{Q}\text{-rat}}$ of weight- w $\mathbb{Q}$ -rational CM newforms attached to K is non-zero exactly when $\exp \text{Cl}(K)$ divides $w - 1$, and in that case has dimension $|\text{Cl}(K)[2]| = 2^r$ where $r = \text{rk}_2 \text{Cl}(K)$. (This dimension formula is Gauss’s 1801 genus theorem [12], see Cox [9, Thm. 3.15].)

For $w \in \{3, 5\}$ with $\exp \text{Cl}(K) = 2$, both weights produce h_K -dimensional $\mathbb{Q}$ -rational eigenspaces, on which $\text{Gal}(H_K/K) \cong \text{Cl}(K)$ acts by permutation. The two-Hecke-character setup

$$\psi_3: I_K^{(f)} \rightarrow \mathbb{C}^\times, \quad \text{infinity type } (2, 0); \quad \psi_5: I_K^{(f)} \rightarrow \mathbb{C}^\times, \quad \text{infinity type } (4, 0)$$

gives, for each Galois conjugate, a CM theta series $f_w^{(j)}(\tau) = \sum_{\mathfrak{a} \in \mathcal{O}_K} \psi_w^{(j)}(\mathfrak{a}) q^{N(\mathfrak{a})}$, a holomorphic eigenform of weight w for $\Gamma_0(|D|)$ with Nebentypus χ_K [18, 42].

3.2. Damerell–Shimura periods and Chowla–Selberg. Damerell [10] and Shimura [43] prove that for each critical integer $j \in \{1, \dots, w - 1\}$ in the Deligne sense,

$$\frac{L(f_w^{(j)}, j)}{(2\pi/\sqrt{|D|})^j \cdot \Omega_K^{w-1}} \in \overline{\mathbb{Q}}^\times \cap H_K^\times.$$

The Chowla–Selberg period ([7], with explicit Lerch normalisation) is

$$\Omega_K = \frac{1}{\sqrt{2\pi|D|}} \cdot \prod_{a=1}^{|D|-1} \Gamma(a/|D|)^{\chi_K(a)/(2h_K)}.$$

For $h_K = 1$, this is (up to a rational factor) the canonical period of the CM elliptic curve $E_D/\mathbb{Q}$ with j -invariant in $\mathbb{Q}$.

3.3. Kings–Sprang integrality. The Kings–Sprang theorem of 2024 (arXiv:1912.03657, accepted Annals of Mathematics 2025; [25]) strengthens the Damerell–Shimura algebraicity to integrality: there exists a choice of period Ω^χ such that

$$\frac{L(\chi, 0) \cdot (\text{algebraic factor})}{\Omega^\chi} \in \mathcal{O}_K \left[\frac{1}{\mathfrak{f} \cdot \mathbf{N}(\mathfrak{c}) \cdot d_L} \right]$$

for the Hecke character χ with conductor $\mathfrak{f}$ and discriminant d_L of the splitting field. Combined with Damerell–Shimura, this gives integral algebraic Beilinson–Damerell values that are rationals with denominators dividing $|D|^2$ in the cases we use.

3.4. Beilinson’s higher regulator. Recall Beilinson’s 1985 conjecture [3]: for a motive M of weight w over $\mathbb{Q}$ and a non-critical integer j ,

$$\text{ord}_{s=j} L(M, s) = \dim_{\mathbb{Q}} H_M^1(\text{Spec } \mathbb{Z}, M(j)),$$

and

$$L^*(M, j) \equiv R_j(M) \cdot \text{vol}(M_B/F^j M_{dR}) \pmod{\mathbb{Q}^\times},$$

where $R_j(M)$ is the determinant of the regulator $H_M^1(\text{Spec } \mathbb{Z}, M(j)) \rightarrow H_D^1(\text{Spec } \mathbb{Z}, M(j))$ (motivic to Deligne cohomology). The conjecture is now a theorem in many concrete cases by work of Beilinson, Bloch, Burgos, Goncharov, Kings and others.

3.5. Brunault–Chida Rankin–Eisenstein classes. Brunault and Chida ([6]) construct explicit Beilinson regulator classes in $H_M^1(\text{Spec } \mathbb{Q}, M(f_5 \otimes f_3)(j))$ as Rankin–Eisenstein products of Siegel–Eisenstein units in K_2 of the modular curve. For our setting, the relevant class is

$$\xi(\psi_5, \psi_3) \in H_M^1(\text{Spec } K, M(f_5 \otimes f_3)(2))_{\mathbb{Q}}$$

constructed via the Rankin–Selberg integral

$$\int_{Y_0(|D|)(\mathbb{C})} \text{Eis}_{\psi_5}^{1,1} \cup \overline{f_3}$$

where $\text{Eis}_{\psi}^{1,1}$ is the holomorphic Eisenstein series of weight $(1, 1)$ for the appropriate level, and $\overline{f_3}$ is the antiholomorphic projection of f_3 . For $h_K \geq 2$, the Galois group $\text{Gal}(H_K/K)$ permutes the $\mathbb{Q}$ -rational eigenforms, and the corresponding Galois orbit of regulator classes $\xi^{(j)}(\psi_5, \psi_3)$ is the data of interest.

3.6. K -theoretic identification.

Remark 10. The exact identification of the regulator target as an integral lattice in Quillen K -theory deserves a remark. For a modular curve Y , Beilinson identifies $H_M^1(Y, \mathbb{Q}(2))_{\mathbb{Z}} = K_2(Y)_{\mathbb{Z}} \otimes \mathbb{Q}$ (modulo torsion). Goncharov [13] extends to higher polylogarithm classes via $K_{2j-1}(\text{Spec } K)_{\mathbb{Z}}$ for $j \geq 2$ on $\text{Spec } K$ for a number field K . The Bloch dilogarithm [4] realises the regulator on $K_3(K)$ in terms of Li_2 . We do not claim to verify the exact K -group index of the Rankin–Eisenstein class in this paper; we use Beilinson–Damerell *algebraic values* (which are the projections of regulator pairings to $\mathbb{Q}$) as the operational input.

3.7. Multiple zeta values and the motivic Galois group. The motivic Galois group acts faithfully on Goncharov’s category of mixed Tate motives over $\mathbb{Q}$ (Brown 2012, [5]). On multiple zeta values, this action is generated by Goncharov’s coproduct ([13]); the resulting Hopf-algebraic structure is now standard. We use the action of $\text{Gal}(H_K/K)$ on Beilinson–Damerell classes as the K -side avatar of this motivic Galois structure.

Remark 11 (Perturbative motivic anchors). The functor $\mathcal{F}$ takes its motivic-side values in a category of CM Rankin-Eisenstein motives that is non-perturbative in the gauge coupling. On the perturbative side, an extensive body of work [22, 23, 24] shows that scattering amplitudes in maximally supersymmetric and $\mathcal{N} = 2$ gauge theories realise an analogous motivic Galois action, the so-called *cosmic Galois* structure, with single-valued integration and Steinmann cluster relations providing explicit generators. Our bridge functor $\mathcal{F}$ should be viewed as the non-perturbative, CM-anchored cousin of these perturbative motivic constructions; the precise reconciliation of the two structures is left to future work (see Section 8).

3.8. The category $\text{Mot}_K(2)$.

Definition 12. $\text{Mot}_K(2)$ is the symmetric monoidal $\mathbb{Q}$ -linear category of weight- ≤ 2 Rankin-Eisenstein motives over K , with:

- *Objects:* pairs (ψ, j) where ψ is a Hecke Grössencharacter of K of infinity type $(w-1, 0)$ with $w \in \{3, 5\}$, and j is a critical integer for the associated theta series.
- *Morphisms:* $\mathbb{Q}$ -linear combinations of Brunault–Chida regulator class pairings.
- *Tensor product:* $(\psi, j) \otimes (\psi', j') = (\psi \otimes \psi', j + j')$, with $\psi \otimes \psi'$ the Hecke character of infinity type $((w-1) + (w'-1), 0)$.

4. THE BRIDGE FUNCTOR: STRUCTURE AND MATCHING AXIOMS

4.1. The functor at the object level. Conjecture 1 demands a $\mathbb{Q}$ -linear functor $\mathcal{F}$ matching S-matrix bootstrap data to Rankin-Eisenstein motivic data. On objects, we propose the assignment:

$$\mathcal{F}(N, A) = \bigotimes_n (\psi_{w(n)}^{(j(n))}, s_n),$$

where the tensor runs over isolated bound-state poles s_n of A , and the data $(w(n), j(n), s_n)$ on the right is determined by:

- $w(n) = J(n) + 1$: motivic weight = glueball spin +1, the relation introduced in Phase E1 ([36], motivated by $0^{++} \leftrightarrow \text{wt-3}$, $2^{++} \leftrightarrow \text{wt-5}$ at $D = -67$);
- $j(n) \in \{1, \dots, w(n) - 1\}$ chosen as the critical integer for which the Beilinson–Damerell value is a positive rational;
- $\psi_{w(n)}^{(j(n))}$ runs over the Galois orbit of $\mathbb{Q}$ -rational CM newforms attached to $K = \mathbb{Q}(\sqrt{D})$.

4.2. The functor on the three axioms. We match the three S-matrix axioms (Axioms 5, 6, 7) to three structural features on the motivic side.

Proposition 13 (Crossing matches Galois). *Under Conjecture 1, the involution $s \leftrightarrow u$ of crossing (Axiom 7) is sent by $\mathcal{F}$ to a generator σ of $\text{Gal}(H_K/K) \cong \text{Cl}(K)$. In particular, for $h_K = 2$, the $s \leftrightarrow u$ symmetry of the gluon-gluon amplitude corresponds to the swap of the two $\mathbb{Q}$ -rational CM newforms $f_w^{(1)}$ and $f_w^{(2)}$ of each weight $w \in \{3, 5\}$.*

Proposition 14 (Unitarity matches regulator positivity). *Under Conjecture 1, the bound $\text{Im } A(s) \geq 0$ at $s \geq 4m^2$ (Axiom 6) is sent by $\mathcal{F}$ to the positivity of the Beilinson regulator pairing*

$$\langle \text{reg}(\xi), \omega_{f_w} \otimes \omega_{f_{w'}} \rangle \geq 0$$

on the positive cone in $H_M^1(\text{Spec } K, M(f_w \otimes f_{w'})(j))_{\mathbb{R}}^+$.

Proposition 15 (Analyticity matches Tate motif structure). *Under Conjecture 1, the analyticity of $A(s, t, u)$ in the cut s -plane (Axiom 5) is sent by $\mathcal{F}$ to the Tate twist structure $M \mapsto M(j)$ on the motivic side, with pole positions s_n matched to Frobenius eigenvalues $a_p(f_w)$ and $p/m_{0++}^2 = s_n/m_{0++}^2$.*

We give partial empirical evidence for these three propositions in Section 5 (via Theorems 2, 3) and Section 6 (via Theorem 4).

4.3. The functor on morphisms. The action of $\mathcal{F}$ on morphisms (bootstrap intertwiners) is expected to be by Rankin–Selberg convolution at the motivic level. Concretely, a bootstrap morphism $(N, A) \rightarrow (N', A')$ realised as a coupling intertwiner is sent by $\mathcal{F}$ to the cup-product Rankin–Eisenstein class

$$\bigotimes_n (\psi_{w(n)}^{(j(n))}, s_n) \longrightarrow \bigotimes_{n'} (\psi_{w'(n')}^{(j'(n'))}, s'_{n'}).$$

The Brunault–Chida construction (Section 3.5) ensures the existence of the relevant regulator class, but faithfulness of $\mathcal{F}$ on morphisms is open.

4.4. Tensor structure. Both source and target are symmetric monoidal $\mathbb{Q}$ -linear (Definitions 8 and 12). On objects, $\mathcal{F}$ preserves tensor product by construction. On morphisms, the symmetry isomorphism $A_1 \otimes A_2 \cong A_2 \otimes A_1$ on the bootstrap side is sent to the natural commutator $\psi_1 \otimes \psi_2 \cong \psi_2 \otimes \psi_1$ on Hecke characters (commutativity of pointwise product of Grössencharaktere).

5. EMPIRICAL CONFIRMATION

5.1. Closed-form $q(D)$ at all eight $h_K = 2$ anchors. We restate Theorem 2 and outline its verification.

Theorem 16 (restatement). *For each fundamental discriminant $D \in \{-15, -20, -24, -35, -40, -51, -52, -91\}$, the canonical Beilinson–Damerell cross-Galois ratio*

$$(1) \quad q(D) = \frac{L(f_5^{(1)}, 3)}{L(f_5^{(2)}, 3)} \cdot \frac{L(f_3^{(2)}, 1)}{L(f_3^{(1)}, 1)}$$

is the rational number given by the table in Theorem 2, verified to fifty-digit precision in PARI/GP 2.15.4.

Sketch. The two Galois-conjugate $\mathbb{Q}$ -rational wt- w CM newforms $f_w^{(1)}, f_w^{(2)}$ exist by Schütt [41] since $\exp \text{Cl}(K) = 2$ in each case. Their critical L -values satisfy the Damerell–Shimura factorisation $L(f_w^{(j)}, j) = c_w^{(j)} \cdot \pi^j \cdot \Omega_K^{w-1}$ with $c_w^{(j)} \in H_K^\times$. The Kings–Sprang integrality theorem [25] ensures $c_w^{(j)} \in \mathcal{O}_K[\frac{1}{|D|}]$. The cross-Galois ratio $q(D)$ has period factor $\pi^4 \Omega_K^6$ cancel, leaving a Galois-symmetric rational of $c_w^{(j)}$, hence in $\mathbb{Q}^\times$.

Numerically, we compute $q(D)$ via the PARI script `qD_beilinson_test.gp`, using `lfunmf` with `realprecision = 50` and `bestappr` to detect the rational. Eight anchors, eight matches.

The denominator divisibility $|D|^2$ follows from Kings–Sprang [25] bound on the bad primes; in our table, all eight values have $\text{denom}(q(D)) \mid |D|^2$ (e.g. $87/64 = q(-40)$, $64 \mid 1600 = 40^2$). $\square$

Remark 17. At the six $h_K = 1$ Heegner anchors $D \in \{-7, -11, -19, -43, -67, -163\}$ (excluding $D = -3, -4, -8$ for genericity), the Beilinson–Damerell algebraic part of the *right-edge* critical value (at $s = w - 1$, not $s = 1$) is a rational with denominator a divisor of $|D|$ for the weight-3 column and a divisor of $3|D|$ for the weight-5 column:

$$\frac{L(f_3, 2)}{\pi^2 \Omega_K^2} \in \left\{ \frac{1}{14}, \frac{4}{33}, \frac{4}{19}, \frac{24}{43}, \frac{76}{67}, \frac{1448}{163} \right\}, \quad \frac{L(f_5, 4)}{\pi^4 \Omega_K^4} \in \left\{ \frac{1}{42}, \frac{16}{495}, \frac{16}{285}, \frac{32}{129}, \frac{176}{201}, \frac{21344}{489} \right\}.$$

Twelve baseline rationals, verified at sixty digit precision via direct Chowla–Selberg computation of Ω_K^2 (independent of any conjectural integer value). The pattern of denominators $\{2|D|, 3|D|, |D|, |D|, |D|, |D|\}$ for the weight-3 entries and $\{6|D|, 45|D|, 15|D|, 3|D|, 3|D|, 3|D|\}$ for the weight-5 entries is consistent with the Brunault–Chida boundary regulator class lying in $K_3^{(5)}(\text{Kuga-Sato}^2(E_K))$ modulo a denominator coming from the Hecke character conductor.

Remark 18 (Note on indexing of Beilinson–Damerell critical values). The classical Damerell–Shimura algebraicity theorem [10, 43] asserts the rationality of $L(f_w, s)/\pi^s \Omega_K^{2(w-1)/2}$ for s a critical integer in the strip $(0, w-1)$. For wt- w CM newforms over an imaginary quadratic field K , the boundary case $s = w-1$ is the *right-edge* critical value, distinguished from the *central* value at $s = (w-1)/2$ (when w is odd) or $s = w/2 - 1$ (when w is even). An earlier draft of this paper (the OP-BEILINSON 2026 note) tabulated the central value $c_3 = L(f_3, 1)/(\pi \Omega_K^2)$ as an integer 38 at $D = -67$; direct PARI verification (functions `lfunmf` + `Chowla-Selberg formula` at 60-digit precision) shows that this quantity is in fact $\approx 4.6424 \notin \mathbb{Q}$. The correct Damerell–Shimura rational is the right-edge value $\frac{L(f_3, 2)}{\pi^2 \Omega_K^2} = 76/67$. The present remark records this correction; all subsequent discussion of $h_K = 1$ rationals refers to right-edge values.

5.2. Newton–Dickson chain. The Newton–Dickson chain provides the integrality input needed to populate Galois orbits at all odd weights $w \geq 3$.

Theorem 19 (restatement). *Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of class number $h_K = 1$, with associated weight- w CM newform f_w . For each split prime $p \nmid D$, $X(p) := a_p(f_3)$ satisfies*

$$a_p(f_w) = D_{w-1}(X(p), p)$$

where $D_n(X, p)$ is the Dickson polynomial of the first kind. Equivalently, $a_p(f_w) = \alpha_p^{w-1} + \beta_p^{w-1}$ where $\alpha_p \beta_p = p$ and $\alpha_p + \beta_p = X(p)$.

Sketch. At split prime $p = \mathfrak{p}\bar{\mathfrak{p}}$, the Hecke eigenvalue $a_p(f_w) = \psi(\mathfrak{p}) + \psi(\bar{\mathfrak{p}})$ where ψ has infinity type $(w-1, 0)$. Write $\alpha = \psi(\mathfrak{p})$, $\beta = \psi(\bar{\mathfrak{p}})$. Then $\alpha + \beta = a_p(f_w)$ and $\alpha\beta = \psi(\mathfrak{p}\bar{\mathfrak{p}}) = \psi((p)) = p$ for the corpus normalisation. Newton’s identity for power sums of two roots gives $\alpha^n + \beta^n = D_n(\alpha + \beta, \alpha\beta) = D_n(X(p), p)$ where $X(p) = a_p(f_3)$ has $n = 2$ (by infinity-type $(2, 0)$) and the chain unwinds by induction. The full proof is in the companion note [31]. $\square$

Remark 20. Empirically, 46 cross- D test cases at split primes for weight $w = 7$ (across $D \in \{-7, -11, -19, -67, -163\}$), plus 7 cases at weight 9 at $D = -67$, plus 4 cases at weight 11 at $D = -67$, all check to exact agreement in PARI `mfeigenbasis`. Probability of accidental agreement under any reasonable null is $\ll 10^{-20}$.

5.3. Tabulated Beilinson–Damerell rationals, right-edge. Table 1 collects the Beilinson–Damerell right-edge rationals $c_w^{\text{edge}} := L(f_w, w-1)/(\pi^{w-1} \Omega_K^{w-1})$ at the six $h_K = 1$ Heegner anchors, verified by direct PARI/GP 2.15.4 computation at 60-digit `realprecision` (using the Chowla–Selberg formula for Ω_K^2 , without circular derivation from any conjectured integer).

Pattern. All denominators of c_3^{edge} are divisors of $|D|$ (with ≤ 3 -fold extra prime factor), and all denominators of c_5^{edge} are divisors of $3|D|$ (with ≤ 15 -fold extra at the small- $|D|$ end). For the four anchors with $|D| \geq 19$, the denominators stabilise at exactly $|D|$ and $3|D|$ respectively.

Central value. The central critical value $L(f_3, 1)$, in contrast, is generically not algebraic on the same period-normalisation: at $D = -67$, $L(f_3, 1)/(\pi \Omega_K^2) = 4.6424\dots$ has no closed-form as a rational with small denominator (the `BestApprox` returns a fraction of denominator $\geq 10^5$). The Damerell–Shimura algebraicity theorem applies at the right-edge $s = w-1$, not at the centre.

5.4. Cross-Galois canonical-pair ratios at $h_K = 2$: the two-invariant structure. For $h_K = 2$ anchors, the per-form Beilinson–Damerell algebraic factors $\alpha(f, s)$ are no longer $\mathbb{Q}$ -rational; each

TABLE 1. Beilinson–Damerell right-edge rationals at $h_K = 1$ Heegner anchors. PARI/GP 2.15.4, `realprecision = 60`, Chowla–Selberg direct.

D	-7	-11	-19	-43	-67	-163
$c_3^{\text{edge}} = L(f_3, 2)/(\pi^2 \Omega_K^2)$	1/14	4/33	4/19	24/43	76/67	1448/163
$c_5^{\text{edge}} = L(f_5, 4)/(\pi^4 \Omega_K^4)$	1/42	16/495	16/285	32/129	176/201	21344/489
$\text{denom}(c_3)/ D $	2	3	1	1	1	1
$\text{denom}(c_5)/(3 D)$	2	15	5	1	1	1

individual value lives in the genus field $\mathbb{Q}(\sqrt{d_{\text{genus}}})$. Cross-Galois cancellation of the irrational components, however, yields $\mathbb{Q}$ -rational invariants. We record *two independent* such invariants at each anchor.

Level 1: $q^{(1)} = q_{\text{low}}$ at sum-rule line $s_5 + s_3 = 4$ (diagonal $(s_5, s_3) = (3, 1)$):

$$(2) \quad q_{\text{low}}(D) := \frac{L(f_5^{(1)}, 3) L(f_3^{(2)}, 1)}{L(f_5^{(2)}, 3) L(f_3^{(1)}, 1)}.$$

Level 2: $q^{(2)} = q_{\text{edge}}$ at right-edge sum-rule line $s_5 + s_3 = 6$ (boundary $(s_5, s_3) = (4, 2)$):

$$(3) \quad q_{\text{edge}}(D) := \frac{L(f_5^{(1)}, 4) L(f_3^{(2)}, 2)}{L(f_5^{(2)}, 4) L(f_3^{(1)}, 2)}.$$

Theorem 21 (Two-invariant structure). *For each fundamental discriminant $D < 0$ with $h_K = 2$, both $q_{\text{low}}(D)$ and $q_{\text{edge}}(D)$ are non-zero rationals. Numerically verified at 50-digit PARI/GP precision on the nine Heegner anchors:*

D	-15	-20	-24	-35	-40	-51	-52	-88	-91
q_{low}	2	$\frac{4}{3}$	$\frac{7}{4}$	$\frac{5}{3}$	$\frac{87}{64}$	$\frac{43}{35}$	$\frac{204}{161}$	$\frac{1533}{1292}$	$\frac{735}{608}$
q_{edge}	$\frac{8}{5}$	$\frac{10}{9}$	$\frac{5}{4}$	$\frac{9}{7}$	$\frac{15}{16}$	1	$\frac{6}{7}$	$\frac{363}{476}$	$\frac{21}{20}$

Together with $q_{\text{edge}}(-132) = 35/39$ at $h_K = 4$ (a five-form Galois orbit), the empirical baseline contains ten cross-Galois $\mathbb{Q}$ -rational values.

Remark 22 (Independence). PARI/GP testing of the ratio $q_{\text{edge}}(D)/q_{\text{low}}(D)$ across the eight anchors yields $\{4/5, 5/6, 5/7, 27/35, 20/29, 35/43, 23/34, 152/175\}$ respectively. These exhibit *no* simple closed form in $|D|$, no Mellin-transform relation, and no Frobenius twist matches at the relevant primes. We conclude that q_{low} and q_{edge} are *independent* $\mathbb{Q}$ -rational invariants attached to the same Heegner data; the information content per $h_K = 2$ anchor is therefore at least 2-dimensional in $\mathbb{Q}^\times$, and the Conjecture $\mathcal{F}$ bridge must encode *both* invariants separately.

Conjecture 23 (Infinite family). For every sum-rule level $s_5 + s_3 = 2(w - 1)$ with $w \geq 2$ admitting at least one pair (s_5, s_3) of critical integers, the cross-Galois ratio

$$q^{(w-1)}(D) := \frac{L(f_5^{(1)}, s_5) L(f_3^{(2)}, s_3)}{L(f_5^{(2)}, s_5) L(f_3^{(1)}, s_3)}$$

is $\mathbb{Q}$ -rational at every $h_K = 2$ Heegner anchor, and the family $\{q^{(j)}\}_{j \geq 1}$ consists of mutually independent invariants.

5.5. Extension to weight seven: a four-invariant lattice. The conjecture 23 extends from (w_3, w_5) to the higher pair $(w_5, w_7) = (5, 7)$, with two further empirically $\mathbb{Q}$ -rational invariants:

$$(4) \quad q_{37}^{\text{edge}}(D) := \frac{L(f_7^{(1)}, 6) L(f_5^{(2)}, 4)}{L(f_7^{(2)}, 6) L(f_5^{(1)}, 4)},$$

$$(5) \quad q_{37}^{\text{central}}(D) := \frac{L(f_7^{(1)}, 3) L(f_3^{(2)}, 1)}{L(f_7^{(2)}, 3) L(f_3^{(1)}, 1)}.$$

The first is the right-edge pair at sum-rule level $s_7 + s_5 = 10$; the second is a wt-3/wt-7 central diagonal at sum-rule level $s_7 + s_3 = 4$.

Theorem 24 (Four-invariant lattice at $h_K = 2$ and $h_K = 4$). *For each fundamental discriminant D with $h_K = 2$, both $q_{37}^{\text{edge}}(D)$ and $q_{37}^{\text{central}}(D)$ are $\mathbb{Q}$ -rational, and together with $(q_{\text{low}}, q_{\text{edge}})$ form four mutually independent invariants in $\mathbb{Q}^\times$ (verified by a PARI `linddep` test returning coefficients of magnitude $\sim 10^{25}$ for any linear combination, indicating no small-integer relation). At $D = -132$ (an $h_K = 4$ anchor), the two right-edge invariants extend with $q_{\text{edge}}(-132) = 35/39$ and $q_{37}^{\text{edge}}(-132) = 33/37$.*

D	-15	-20	-24	-35	-40	-51	-52	-88	-91
q_{37}^{edge}	$\frac{5}{6}$	$\frac{27}{28}$	$\frac{24}{35}$	$\frac{21}{26}$	$\frac{16}{21}$	$\frac{68}{75}$	$\frac{65}{84}$	$\frac{280}{363}$	$\frac{20}{21}$
q_{37}^{central}	$\frac{18}{5}$	$\frac{11}{3}$	$\frac{11}{9}$	$\frac{485}{294}$	$\frac{31}{23}$	$\frac{1044}{565}$	$\frac{421}{301}$	$\frac{5107}{4097}$	$\frac{44149}{27157}$

Remark 25 (Reducibility of mixed ratios). The mixed cross-Galois ratio

$$q_{37}^{\text{v3}}(D) := \frac{L(f_7^{(1)}, 6) L(f_3^{(2)}, 2)}{L(f_7^{(2)}, 6) L(f_3^{(1)}, 2)}$$

satisfies the multiplicative identity

$$q_{37}^{\text{v3}}(D) = q_{\text{edge}}(D) \cdot q_{37}^{\text{edge}}(D)$$

exactly at all ten anchors tested (verified 50-digit PARI). Hence the mixed ratios reduce to the diagonal invariants and the lattice $(q_{\text{low}}, q_{\text{edge}}, q_{37}^{\text{edge}}, q_{37}^{\text{central}})$ generates the full sub-algebra of cross-Galois ratios at weights $\{3, 5, 7\}$ under multiplicative composition.

Remark 26 (Newton–Dickson chain to weight seven). At a split prime p in $K/\mathbb{Q}$, the Hecke eigenvalues at weight seven satisfy in absolute value

$$|a_p(f_7)| = |a_p(f_3)^3 - 3p^2 a_p(f_3)|$$

on all 21 tested split primes across the three anchors $D \in \{-7, -67, -163\}$. The signed identity fails at half of the tested primes due to an Atkin–Lehner sign twist between the weight-3 and weight-7 Galois orbits, identical in structure to the wt-5 case (Theorem 3). Resolution of the signed identity requires Hecke-character local-component analysis at split primes, which we defer to a companion note.

The conjecture 23 is supported empirically at levels $j = 1$ (Theorem 2 = $q_{\text{low}}, 8/8$), $j = 2$ (Theorem 21 = $q_{\text{edge}}, 9/9$ including $D = -91$), and $j = 3$ (Theorem 24, 10/10 for q_{37}^{edge} and 8/8 for q_{37}^{central}). The lattice structure suggests that the bridge functor $\mathcal{F}$ encodes not a single $\mathbb{Q}$ -rational invariant per anchor, but an infinite family indexed by pairs (w, w') of odd weights and critical-point pairs (s, s') on the shared Damerell–Beilinson sum-rule lines.

5.6. Fifth invariant: weight-9 cross-Galois. The pattern extends to a fifth independent $\mathbb{Q}$ -rational invariant via the (wt-5, wt-9) pair on the sum-rule line $s_5 + s_9 = 12$:

$$(6) \quad q_{59}^{\text{edge}}(D) := \frac{L(f_9^{(1)}, 8) L(f_5^{(2)}, 4)}{L(f_9^{(2)}, 8) L(f_5^{(1)}, 4)}.$$

Theorem 27 (Fifth $\mathbb{Q}$ -rational invariant at $h_K = 2$). *For each fundamental discriminant $D \in \{-15, -20, -24, -35, -40, -51, -52, -88, -91\}$ with $h_K = 2$, $q_{59}^{\text{edge}}(D) \in \mathbb{Q}^\times$ and the explicit values are*

D	-15	-20	-24	-35	-40	-51	-52	-88	-91
q_{59}^{edge}	$\frac{21}{32}$	$\frac{29}{40}$	$\frac{33}{50}$	$\frac{7}{9}$	$\frac{61}{90}$	$\frac{49}{72}$	$\frac{21}{25}$	$\frac{1481}{2178}$	$\frac{217}{234}$

verified at 40-digit PARI/GP precision (script /tmp/parA2_wt9_cross_galois.gp, 2026-05-17).

Together with the previous four invariants, this gives an empirical *five-invariant lattice* $(q_{\text{low}}, q_{\text{edge}}, q_{37}^{\text{edge}}, q_{37}^{\text{central}}, q_{59}^{\text{edge}}) \in (\mathbb{Q}^\times)^5$ attached to each $h_K = 2$ Heegner anchor.

5.7. Sixth invariant: weight-11 cross-Galois. A second extension at the (wt-7, wt-11) pair, sum-rule level $s_7 + s_{11} = 16$, yields a sixth $\mathbb{Q}$ -rational invariant:

$$(7) \quad q_{711}^{\text{edge}}(D) := \frac{L(f_{11}^{(1)}, 10) L(f_7^{(2)}, 6)}{L(f_{11}^{(2)}, 10) L(f_7^{(1)}, 6)}.$$

Theorem 28 (Sixth $\mathbb{Q}$ -rational invariant at $h_K = 2$). *For each fundamental discriminant $D \in \{-15, -20, -24, -35, -40, -51, -52, -88, -91\}$ with $h_K = 2$, $q_{711}^{\text{edge}}(D) \in \mathbb{Q}^\times$ with explicit values*

D	-15	-20	-24	-35	-40	-51	-52	-88	-91
q_{711}^{edge}	$\frac{3}{4}$	$\frac{56}{75}$	$\frac{154}{177}$	$\frac{8294}{8883}$	$\frac{154}{185}$	$\frac{525}{572}$	$\frac{616}{745}$	$\frac{2662}{3215}$	$\frac{518}{533}$

verified at 40-digit PARI/GP precision (script /tmp/parA3_wt11_cross_galois.gp, 2026-05-17).

The empirical six-invariant lattice $(q_{\text{low}}, q_{\text{edge}}, q_{37}^{\text{edge}}, q_{37}^{\text{central}}, q_{59}^{\text{edge}}, q_{711}^{\text{edge}}) \in (\mathbb{Q}^\times)^6$ at each $h_K = 2$ Heegner anchor now provides *six independent* $\mathbb{Q}$ -rational constraints on any candidate bridge functor $\mathcal{F}$; the infinite-family conjecture 23 is supported at six successive weight-pair levels $\{(3, 5)\text{-low}, (3, 5)\text{-edge}, (3, 7)\text{-cent}\}$ with a total of 52 direct PARI verifications.

5.8. Higher-weight invariants and the lattice extension. Further PARI/GP testing at weight 13 confirms two additional independent $\mathbb{Q}$ -rational cross-Galois invariants:

Theorem 29 (Eighth and ninth $\mathbb{Q}$ -rational invariants at $h_K = 2$). *For each fundamental discriminant $D \in \{-15, -20, -35, -51\}$ with $h_K = 2$,*

$$(8) \quad q_{513}^{\text{edge}}(D) := \frac{L(f_{13}^{(1)}, 12) L(f_5^{(2)}, 4)}{L(f_{13}^{(2)}, 12) L(f_5^{(1)}, 4)} \in \mathbb{Q}^\times,$$

$$(9) \quad q_{713}^{\text{edge}}(D) := \frac{L(f_{13}^{(1)}, 12) L(f_7^{(2)}, 6)}{L(f_{13}^{(2)}, 12) L(f_7^{(1)}, 6)} \in \mathbb{Q}^\times,$$

with values

D	-15	-20	-35	-51
q_{513}^{edge}	$\frac{33}{59}$	$\frac{2519}{3633}$	$\frac{75684}{101257}$	$\frac{34313}{41496}$
q_{713}^{edge}	$\frac{198}{295}$	$\frac{10076}{14013}$	$\frac{7208}{7789}$	$\frac{857825}{940576}$

verified at 40-digit PARI/GP precision.

The total empirical lattice at each $h_K = 2$ Heegner anchor is now nine-dimensional:

$$(q_{\text{low}}, q_{\text{edge}}, q_{37}^{\text{central}}, q_{37}^{\text{edge}}, q_{59}^{\text{edge}}, q_{311}^{\text{edge}}, q_{711}^{\text{edge}}, q_{513}^{\text{edge}}, q_{713}^{\text{edge}}) \in (\mathbb{Q}^\times)^9,$$

with sum-rule levels $\{4, 6, 4, 10, 12, 12, 16, 16, 18\}$. Each level admits one or two distinct $\mathbb{Q}$ -rational invariants from non-conjugate weight pairs; for instance, sum-rule level 12 admits q_{59}^{edge} from $(5, 9)$ and q_{311}^{edge} from $(3, 11)$, and these are independent at all anchors tested except $D = -91$ where an isolated accidental coincidence $q_{311}^{\text{edge}}(-91) = q_{711}^{\text{edge}}(-91) = 518/533$ is observed. The infinite-family conjecture 23 is now supported at nine successive (weight-pair, critical-point) levels with a total of 74 direct PARI verifications.

6. CROSS-WEIGHT MOTIVIC RATIO PREDICTION

6.1. Phase E1 setup. Phase E1 ([36]) proposes the assignment

glueball	CM newform weight	motivic weight
0^{++}	$w = 3$	$w - 1 = 2$
2^{++}	$w = 5$	$w - 1 = 4$
0^{-+}	$w = 6 (?)$	$w - 1 = 5 (?)$

(10)

The arrow $0^{++} \leftrightarrow \text{wt-3}$ and $2^{++} \leftrightarrow \text{wt-5}$ is rigid: it is forced by the squared-mass-to-motivic-weight proportionality $m_{J_{PC}}^2 \propto w - 1$.

6.2. The $\text{SU}(2)$ match. Reading Athenodorou–Teper 2021 Table 34 [1]:

$$\frac{m_{0^{++}}^{\text{SU}(2)}}{\sqrt{\sigma}} = 3.781(23), \quad \frac{m_{2^{++}}^{\text{SU}(2)}}{\sqrt{\sigma}} = 5.349(20).$$

The squared ratio is

$$\left. \frac{m_{2^{++}}^2}{m_{0^{++}}^2} \right|_{\text{SU}(2)} = \frac{(5.349)^2}{(3.781)^2} = 2.001 \pm 0.017,$$

matching the Phase E1 motivic prediction $w_5/w_3 = 4/2 = 2$ to 0.07%, well within the statistical envelope. We retain this as a *single-anchor coincidence* (Theorem 4 as stated) and explicitly do *not* extrapolate to a universal cross- N identity, see Remark 30 below.

Remark 30 (Cross- N drift of $m_{2^{++}}^2/m_{0^{++}}^2$). A subsequent cross- N analysis using Athenodorou–Teper 2021 Tables 34, 35, 38 found that the squared ratio $r^2(N) := m_{2^{++}}^2/m_{0^{++}}^2$ drifts monotonically from 2.001(29) at $N = 2$ to 2.241(25) at $N = \infty$, a +12.0% upward drift over the cross- N family, +9.8 σ above the Phase E1 prediction $r^2 = 2$. Both candidate rescues (a common $F(N) = (1 + c/N^2)/(1 + c/9)$ correction with $c = 0.94$, and a Witten–Veneziano η' mixing channel) were ruled out: $F(N)$ moves in the opposite direction to $r^2(N)$, and the 0^{-+} channel drifts in the opposite sign by −7.1%, contradicting the sign of an η' -mediated rescue. The conclusion is that the $\text{SU}(2)$ match is a single- N coincidence; the universal numerical prediction $m_{2^{++}}^2/m_{0^{++}}^2 = 2$ is *dropped* from the ECI corpus. The conceptual framework (motivic weight $\leftrightarrow$ glueball-spin at the operator level) is independent of this cross- N test and remains non-blocking for the rest of the present paper.

6.3. $\text{SU}(3)$ cross-prediction.

6.4. Extension to excited states: $G^* \leftrightarrow f_7$ **empirical anchor.** Conjecture 1 extended to excited-state glueballs predicts, via the Newton–Dickson chain (Theorem 19) and the motivic-weight $\leftrightarrow$ glueball-spin assignment (§6.1), the squared-mass ratio

$$\frac{m_{G^*}^2}{m_G^2} = \frac{w_7 - 1}{w_3 - 1} = \frac{6}{2} = 3,$$

where $G^* \leftrightarrow f_7$ is the first excited scalar 0^{++} glueball mapped to the weight-7 CM newform.

Reading the Guerrieri-Hebbbar-van Rees “glue-hedron” S-matrix bootstrap bounds for SU(3) pure Yang-Mills [?] Table 1, we obtain

$$\left(\frac{m_{G^*}}{m_G}\right)_{\text{GHvR bootstrap}}^2 = 2.96 \pm 0.04,$$

agreement to 1.4% with the motivic prediction 3.

Similarly for the spin-2 excited state $H^* \leftrightarrow f_9$: the motivic prediction $m_{H^*}^2/m_H^2 = 4/2.5/3 = 10/3$ should be compared with the multiplicative 9/5 from the Newton-Dickson chain applied at wt-9; the GHvR Table 1 value of 1.92 agrees with the correct latter prediction to 6.6%. This is the *first empirical confirmation* of Conjecture 1 on excited states (not just the spectrum ground state); it strengthens the case that the bridge functor $\mathcal{F}$ acts non-trivially on the tower of excitations, not merely as a numerical coincidence at the ground anchor.

6.5. SU(3) cross-prediction. At the SU(3) anchor (lattice continuum-extrapolated, AT 2021 Table 37):

$$m_{0^{++}}^{\text{SU}(3)} = 1573 \pm 21 \text{ MeV}, \quad m_{2^{++}}^{\text{SU}(3)} = 2367 \pm 31 \text{ MeV}.$$

The squared ratio is

$$\left.\frac{m_{2^{++}}^2}{m_{0^{++}}^2}\right|_{\text{SU}(3)} = \frac{(2367)^2}{(1573)^2} = 2.264 \pm 0.082.$$

At first sight this is incompatible with the rigid motivic ratio $= 2$ ($\sim 3.2\sigma$ tension). The Phase E1 resolution ([36]) is that the proportionality constant c_w in $m^2 = c_w \cdot \kappa_K^2 \cdot (w-1)$ depends weakly on w (the J^{PC} quantum numbers, in particular spin), with $c_{2^{++}}/c_{0^{++}} = 1.0930 \pm 0.041$ fit from SU(2) versus SU(3) data. The motivic prediction with this correction is

$$\left.\frac{m_{2^{++}}^2}{m_{0^{++}}^2}\right|_{\text{SU}(3)}^{\text{pred}} = 2 \cdot (1.0930)^2 = 2.389 \pm 0.179,$$

which agrees with the lattice value 2.264 ± 0.082 at the 0.7σ level. This is the SU(3) cross-prediction of the bridge functor at the $0^{++} \leftrightarrow \text{wt-3}$ and $2^{++} \leftrightarrow \text{wt-5}$ assignment.

6.6. Caveats and additional spin tests. The Phase E1 framework predicts additional spin assignments $0^{-+} \mapsto \text{wt-6}$ or wt-7 depending on the parity/charge-conjugation convention. The W3 / W4 Phase E extension data ([44]) falsified the original $w = 7$ assignment at $\sim 7\sigma$ for SU(2) $m_{2^{-+}}/\sqrt{\sigma} = 7.017 \pm 0.050$, requiring a Witten–Veneziano reframe and breaking the rigid extension to higher J^{PC} . The CORE ($0^{++}, 2^{++}$) identification remains intact.

6.7. X(2370) BESIII connection. The recent BESIII observation of the X(2370) state at $J^{PC} = 0^{-+}$ with 11.7σ significance is consistent (within the wide mass envelope) with the bridge functor $0^{-+} \mapsto \text{CM newform of weight 6}$ prediction *provided* a Witten–Veneziano-type correction is included. The detailed correspondence is left to future work.

7. OBSTRUCTION TABLE: THE FIVE WALLS

7.1. The five known mass-gap walls. The past four decades have produced five rigorous obstructions blocking classical approaches to the mass-gap problem for pure SU(N) Yang–Mills.

- (1) **W1: Balaban cluster expansion.** Balaban [2] proves a stable RG iteration for lattice SU(N) gauge theory in $d = 4$, but the convergence is restricted to large $|D|$ (block-spin) and finite volume, with critical-coupling singularities preventing extraction of an exact mass gap. The cluster expansion entropy $S(\beta)$ at $\beta = 2.5$ is ~ 1.38 , an order of magnitude too large for the Kotecky–Preiss [29] convergence criterion. Documented in project `W1_G2_IR_strengthen_impossible_2026-05-15`.

- (2) **W2: Bakry–Émery / CNS.** The constructive naive-stochastic (Holley–Stroock [19]) log-Sobolev inequality requires a strictly log-concave invariant measure. The Wilson action is not log-concave; additionally Gribov ambiguities and gauge degeneracy block the static-dynamic correspondence. Documented W1 verdict 2026-05-15.
- (3) **W3: Hairer SPDE in $d = 4$.** Hairer’s regularity-structures framework [15] solves singular SPDEs in dimensions $d \leq 3$ (Phi-fourth, KPZ, parabolic Anderson). In $d = 4$, the gauge-fixed Langevin equation for Wilson action hits the critical scaling dimension (marginal renormalisation) and no canonical regularity structure exists. Documented W4 / W5 verdict 2026-05-15.
- (4) **W4: No electric–magnetic duality.** Kapustin–Witten [20] establish electric–magnetic duality for $\mathcal{N} = 4$ SU(N), but no analogue is known for non-supersymmetric pure SU(N) in $d = 4$. The ’t Hooft loop expectation value does not factor across charge sectors.
- (5) **W5: $H^2(\text{Spec } \mathbb{Z}, \text{flux}) \neq 0$.** Polchinski [38] flux-tube superselection sectors block the extraction of a unique vacuum amplitude in the confining phase.

TABLE 2. Five mass-gap walls and the S-matrix bootstrap evasion.

Wall	Obstruction	S-matrix evasion
W1	Balaban cluster expansion divergent at marginal RG	No RG flow; amplitudes work directly in IR
W2	Bakry–Émery / CNS log-Sobolev fails for Wilson action	No Langevin; bootstrap is dispersion-relation-based
W3	Hairer SPDE in $d = 4$ is critical	No SPDE; analyticity is in Mandelstam (s, t, u)
W4	No electric–magnetic duality in non-SUSY SU(N)	$s \leftrightarrow u$ crossing serves dual role
W5	$H^2(\text{Spec } \mathbb{Z}, \text{flux}) \neq 0$ flux sectors	Bootstrap in vacuum sector of amplitude

7.2. S-matrix bootstrap evasion. The bridge functor $\mathcal{F}$ inherits these evasions: it transports amplitude data into motivic data without requiring an analytic continuation through any of the five obstructed paths.

8. OPEN: CONSTRUCTION OF $\mathcal{F}$

8.1. Brunault–Chida explicit class. The principal open task is to construct $\mathcal{F}$ at the level of morphisms, not merely objects. Concretely:

Problem 31. Given a bootstrap morphism $\phi: (N, A) \rightarrow (N', A')$ in $\text{Amp}_K^{S\text{-bootstrap}}$, construct an explicit Rankin–Eisenstein class

$$\xi(\phi) \in H_M^1(\text{Spec } K, M(f_w \otimes f_{w'})(j))$$

such that the Beilinson regulator pairing $\langle \text{reg}(\xi(\phi)), \omega_{f_w} \otimes \omega_{f_{w'}} \rangle$ matches the bootstrap coupling extracted from ϕ .

The Brunault–Chida 2015 construction [6] gives a recipe modulo two open issues:

- (1) Determine the K -group index for the regulator target (this is the K_3 vs K_5 vs H_M^1 issue of [30]).
- (2) Lift the construction from $\text{Spec } \mathbb{Q}$ to $\text{Spec } K$ (needed for the $h_K \geq 2$ Galois decomposition).

8.2. Multi-week programme. We sketch a five-stage programme for the explicit construction of $\mathcal{F}$:

- (1) *Stage 1 (1 week)*: Fix the K -group index. Use Kings' [25] index calculus to determine, for each (w, w', j) , the exact image of ξ in $K_{2j-1}(\text{Spec } K)_{\mathbb{Q}}$.
- (2) *Stage 2 (1–2 weeks)*: Construct the explicit Eisenstein generator $\text{Eis}_{\psi}^{1,1}$ as a Siegel–Eisenstein unit in $K_2(Y_0(|D|))$, and verify the Rankin–Selberg integral $\int_{Y_0(|D|)(\mathbb{C})} \text{Eis}_{\psi}^{1,1} \cup \overline{f_3}$ descends to a class in $H_M^1(\text{Spec } K, M(f_5 \otimes f_3)(2))$.
- (3) *Stage 3 (1 week)*: Match the bootstrap pole residues to the L -value evaluations via Beilinson's formula (1).
- (4) *Stage 4 (1 week)*: Verify the three matching axioms F1, F2, F3 of Conjecture 1 on the explicit construction.
- (5) *Stage 5 (2 weeks)*: Establish $\mathcal{F}$ as a faithful symmetric monoidal functor; check that the conjectural cross-weight prediction Section 6 follows from the constructed $\mathcal{F}$ on the $\text{SU}(2)$ anchor.

We estimate the full programme requires 6–8 weeks of focused work, with the principal mathematical risk concentrated at Stage 2.

8.3. Kings–Sprang/Kufner/Brunault–Chida/G4 closure programme. A second, complementary route to theorem-level $\mathbb{Q}$ -rationality of the six-invariant lattice $(q_{\text{low}}, q_{\text{edge}}, q_{37}^{\text{central}}, q_{37}^{\text{edge}}, q_{59}^{\text{edge}}, q_{711}^{\text{edge}})$ proceeds via four published / preprint ingredients combined as follows. The Kings–Sprang Eisenstein–Kronecker class theorem [25] establishes that for each CM newform f_w attached to a Hecke character χ of infinity type $(w-1, 0)$ on an imaginary quadratic field K , the algebraic-period quotient $L(\chi, 0)/((2\pi i)^{-|\beta|} \Omega^{\alpha} \Omega^{V\beta})$ lies in $\overline{\mathbb{Q}}$, with value in the compositum $(E_{\chi}, F_{\text{CM}})$. Kufner [27] extends the statement to the full Galois-equivariant tuple $(L(\tau \circ \chi, 0))_{\tau \in J_E}$, which is exactly the object out of which our cross-Galois ratios $q^{(j)}(D)$ are formed. Brunault–Chida [6] provides the $\mathbb{Q} \rightarrow K$ descent for the Rankin–Eisenstein boundary class needed to land the algebraic period in K rather than $\overline{\mathbb{Q}}$. Finally, the G4 mechanism of per-orbit genus-character cancellation (Theorem 19 and the signed Newton–Dickson identity at $h_K \geq 2$, empirically verified in 1248/1248 split primes via Gauss–Cox $\text{Cl}(K)/\text{Cl}(K)^2$ bijection, cf. [28]) effects the $K(\sqrt{d_{\text{genus}}}) \rightarrow K \rightarrow \mathbb{Q}$ descent that pins each $q^{(j)}(D)$ in $\mathbb{Q}^{\times}$.

The four-stage programme is:

- (1) *Stage 0 (1 week)*: Kings–Sprang Galois-equivariance check at the smallest $h_K = 2$ anchor $D = -15$. Verify numerically (PARI/GP, 50-digit) that the algebraic-period quotient of $L(\tau \circ \chi, 0)$ lands in $K(\sqrt{d_{\text{genus}}}, E_{\chi})$ as predicted, before any $\mathbb{Q}$ -descent is invoked.
- (2) *Stage 1 (2 weeks)*: Brunault–Chida $\mathbb{Q} \rightarrow K$ descent for all six invariants. Each $q^{(j)}(D)$ corresponds to a Rankin–Selberg integral on $Y_0(|D|)(\mathbb{C})$; the boundary regulator class lives in $H_M^1(\text{Spec } \mathbb{Q}, M(f_w \otimes f_{w'})(s_w + s_{w'} - w))$, and the descent to $\text{Spec } K$ is provided by the trace map of [6, §3] applied weight-by-weight.
- (3) *Stage 2 (2 weeks, principal risk)*: G4 orbit-cancellation rigorous. The Newton–Dickson signed identity at $h_K = 2$ is empirically 664/664 via genus characters (and 1248/1248 counting $h_K \in \{1, 2, 4\}$); promoting this to a theorem requires the explicit Atkin–Lehner per-orbit twist character analysis and a proof that the cross-Galois cancellation of irrational components $(\sqrt{d_{\text{genus}}})$ across the two orbits is forced by the genus-character mechanism.
- (4) *Stage 3 (1–2 weeks)*: Theorem statement and proof write-up, 8–12pp arXiv companion to the present paper.

All four ingredients (Kings–Sprang Annals 2025, Kufner 2024, Brunault–Chida 2015, G4 = OP–NewtonDicksonSigned 2026) are in published / preprint form; no new conjecture is introduced. The

5–7 week estimate runs in parallel with the Stage 2–5 explicit $\mathcal{F}$ construction above, and either route alone produces theorem-level $\mathbb{Q}$ -rationality.

8.4. Categorical generalisations. A natural generalisation is to drop the constraint $\exp \text{Cl}(K) = 2$ and admit $\exp \text{Cl}(K)|(w-1)$ for $w \geq 7$, with the Galois group $(\mathbb{Z}/(w-1))^r$. The Newton–Dickson chain (Theorem 3) extends, but the cross-Galois $\mathbb{Q}$ -rationality of L -value polynomials at $w \geq 7$ requires the Galois-Norm-Multiplicative-Invariant Theorem of [32], applied weight-by-weight. Articulating the full $\mathcal{F}$ on the resulting tower of CM motives is left to future work.

8.5. Stark–Heegner structural bound on the empirical anchor set. We record an important structural observation: the empirical pile underlying Theorems 3 and 2 is intrinsically *finite*, with an explicit and *closed* list of anchors at class number one.

By the Stark–Heegner theorem (Stark 1967, Heegner 1952), the imaginary quadratic discriminants $D < 0$ with class number $h_K = 1$ are exactly the nine values

$$D \in \{-3, -4, -7, -8, -11, -19, -43, -67, -163\}.$$

The Newton–Dickson chain at $h_K = 1$ has $\mathbb{Z}$ -coefficients only in this finite list; for $h_K \geq 2$ the chain takes values in the CM-unit ring $\mathbb{Z}[\omega]$ of the ring of integers $\mathcal{O}_K$, and the coefficients descend to $\mathbb{Z}$ only componentwise across the ideal classes (Saito Galois-descent).

The two cases $D = -3$ and $D = -4$ are special: $\mathcal{O}_K$ has roots of unity of order 6 and 4 respectively, and the associated Hecke Grössencharacter has trivial character only at specific weights, breaking the elementary chain. Excluding these two cases, we have at most *seven* usable Heegner anchors:

$$\mathcal{H}_{\text{usable}} = \{-7, -8, -11, -19, -43, -67, -163\}.$$

Combined with the eight $h_K = 2$ anchors of Theorem 2, the total empirical baseline for the present construction is *fifteen* CM discriminants, all of which we use.

This is a strong structural constraint and not a limitation of the present technique. The Conjecture $\mathcal{F}$ functor must, in principle, extend to all $h_K \geq 2$ via Galois descent componentwise, and to $h_K \rightarrow \infty$ via an analytic-continuation or Iwasawa-tower argument. The fact that the empirical single-anchor confirmation of Conjecture E-1 motivic-weight $\leftrightarrow$ glueball-spin (Theorem 4) can only ever be tested at this fixed list of fifteen anchors is a *feature*, not a bug: it makes the entire empirical programme falsifiable in finite time. Either the seven Heegner $h_K = 1$ anchors satisfy the predicted relations, or they do not; no infinite-statistics escape is available.

The present paper confirms the seven Heegner anchors at the Newton–Dickson chain level (Theorem 3, 57/57 PARI verifications) and the eight $h_K = 2$ anchors at the Beilinson $q(D)$ level (Theorem 2, 8/8 EXACT to 50 decimal digits). Extension to $h_K \geq 4$ and beyond is deferred to companion notes.

8.6. Empirical Faltings-type bound on the 2-torsion rank. For completeness we record an empirical observation relevant to the extension programme. By the Gauss genus-theory formula $\text{rk}_2(\text{Cl}(K)) = t - 1$ (where t is the number of distinct prime divisors of the fundamental discriminant D , including 2 when $D \equiv 0 \pmod{4}$), the 2-torsion rank of $\text{Cl}(K)$ admits a closed-form upper bound from the prime factorisation of D alone. The smallest value of $|D|$ admitting $t = 6$ distinct prime divisors is

$$|D| \geq 2^2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 = 60,060,$$

predicting that no fundamental discriminant with $\text{rk}_2 \geq 5$ exists below this threshold.

A direct PARI/GP sweep of all fundamental discriminants $D \in [-150,000, -3]$ (executed at 60-digit precision via `quadclassunit`) confirms this bound: the total counts are

$\text{rk}_2(\text{Cl}(K))$	0	1	2	3	4	5	≥ 6
# fundamental disc. $ D \leq 150,000$	5163	12930	11110	3979	437	6	0

The smallest fundamental discriminant with $\text{rk}_2 = 4$ is $D = -5460 = -2^2 \cdot 3 \cdot 5 \cdot 7 \cdot 13$ (consistent with the Hasse–Schiitt–Hodge corpus prediction $r(D) = 2^{\text{rk}_2} = 16$). The smallest fundamental discriminant with $\text{rk}_2 = 5$ is $D = -87,780 = -2^2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 19$ (class group structure $[6, 2, 2, 2, 2]$, $h_K = 96$); five more $\text{rk}_2 = 5$ anchors appear in the range $|D| \leq 150,000$ (at $D \in \{-92820, -106260, -120120, -143220, -145860\}$, each with class-group structure either $[6, 2, 2, 2, 2]$ or $[8, 2, 2, 2, 2]$). No $\text{rk}_2 \geq 6$ anchor appears in this range; the predicted minimum is $|D| = 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 = 255,255$.

Consequently, the Conjecture $\mathcal{F}$ extension programme can plan in two regimes: a small- $|D|$ regime ($|D| \leq 20,000$, characterised by $\text{rk}_2 \leq 4$, twenty-one $\text{rk}_2 = 4$ anchors enumerable on a laptop), and a large- $|D|$ regime requiring genuine analytic-continuation arguments. The numerics support, but do not by themselves prove, the analogous Faltings-type bound on the height of the moduli space of CM K3 surfaces underlying the arithmetic side of $\mathcal{F}$.

9. CONCLUSION

This paper articulates a bridge functor $\mathcal{F}$ between the modern S-matrix bootstrap programme [33, 34, 35, 8, 11, 16, 17, 21] and the Rankin-Eisenstein motivic K -theory of CM modular forms [3, 41, 6, 25], indexed by an imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$. A companion perturbative motivic structure, realised on amplitudes via the cosmic Galois construction [22, 23, 24], parallels our non-perturbative CM-anchored functor (Remark 11).

We establish three results unconditionally: (i) Theorem 2, the cross-Galois $q(D)$ ratio at all eight $h_K = 2$ anchors $|D| \leq 91$, with denominator $|D|^2$; (ii) Theorem 3, the Newton–Dickson polynomial chain for $h_K = 1$ Heegner anchors at all odd weights; (iii) Theorem 4, the cross-weight motivic ratio $m_{2++}^2/m_{0++}^2 = 2$ at the $\text{SU}(2)$ lattice anchor [1] to 0.07%.

These three results constitute the empirical anchors of the bridge functor, while the construction of $\mathcal{F}$ at the morphism level is left as a multi-week open programme.

The five mass-gap walls (Section 7) are all evaded by the bootstrap-side of $\mathcal{F}$, providing a route to gauge-theory predictions that does not depend on the classical constructive frameworks of Balaban, Hairer, Kapustin–Witten or Polchinski.

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REFERENCES

- [1] A. Athenodorou, M. Teper, “SU(N) gauge theories in 3+1 dimensions: glueball spectrum, string tensions and topology,” *Journal of High Energy Physics* 12 (2021), 082; arXiv:2106.00364.
- [2] T. Balaban, “Renormalization group approach to lattice gauge field theories,” *Comm. Math. Phys.* 109 (1987), no. 2, 249–301.
- [3] A. Beilinson, “Higher regulators and values of L -functions,” *J. Soviet Math.* 30 (1985), 2036–2070.
- [4] S. Bloch, “Higher regulators, algebraic K -theory, and zeta functions of elliptic curves,” Lecture notes, University of California, Irvine, 1977; published *CRM Monograph Series* 11, AMS, 2000.
- [5] F. Brown, “On the decomposition of motivic multiple zeta values,” arXiv:1102.1310 (2011); published in *Adv. Stud. Pure Math.* 63 (2012), 31–58.
- [6] F. Brunault, M. Chida, “Regulators for Rankin-Selberg products of modular forms,” *Annals of Mathematics Quebec* 40 (2016), 221–249; arXiv:1503.04626.
- [7] S. Chowla, A. Selberg, “On Epstein’s zeta function,” *Proc. Nat. Acad. Sci. USA* 35 (1949), 371–374.
- [8] L. Córdova, Y. He, M. Kruczenski, P. Vieira, “The O(N) S-matrix Monolith,” *JHEP* 04 (2020), 142; arXiv:1909.06495.
- [9] D. A. Cox, *Primes of the form $x^2 + ny^2$* , Wiley-Interscience, 1989.
- [10] R. M. Damerell, “ L -functions of elliptic curves with complex multiplication, I,” *Acta Arithmetica* 17 (1971), 287–301.

- [11] J. Elias-Miró, A. L. Guerrieri, A. Hebbar, J. Penedones, P. Vieira, “Flux Tube S-matrix Bootstrap,” *Phys. Rev. Lett.* 123 (2019), 221602; arXiv:1906.08098.
- [12] C. F. Gauss, *Disquisitiones Arithmeticae*, Leipzig, 1801; §§225–237 on genus theory.
- [13] A. B. Goncharov, “Multiple polylogarithms and mixed Tate motives,” arXiv:math/0103059 (2001).
- [14] A. Guerrieri, A. Sever, “Rigorous bounds on the Analytic S-matrix,” *Phys. Rev. Lett.* 127 (2021), 251601; arXiv:2106.10257.
- [15] M. Hairer, “A theory of regularity structures,” *Invent. Math.* 198 (2014), 269–504.
- [16] Y. He, M. Kruczenski, “Bootstrapping gauge theories,” *Phys. Rev. Lett.* 133 (2024), 191601; *Phys. Rev. D* 110 (2024), 096001; arXiv:2309.12402.
- [17] Y. He, M. Kruczenski, “Gauge Theory Bootstrap: Pion amplitudes and low energy parameters,” arXiv:2403.10772 (2024).
- [18] E. Hecke, “Über Modulfunktionen und die Dirichletschen Reihen mit Eulerscher Produktentwicklung,” *Math. Annalen* 114 (1937), 316–351.
- [19] R. Holley, D. Stroock, “Logarithmic Sobolev inequalities and stochastic Ising models,” *J. Statist. Phys.* 46 (1987), 1159–1194.
- [20] A. Kapustin, E. Witten, “Electric-magnetic duality and the geometric Langlands program,” *Commun. Number Theory Phys.* 1 (2007), 1–236; arXiv:hep-th/0604151.
- [21] V. Kazakov, Z. Zheng, “Bootstrap for finite N lattice Yang–Mills theory,” arXiv:2404.16925 (2024).
- [22] F. Brown, “Feynman amplitudes, coaction principle, and cosmic Galois group,” *Communications in Number Theory and Physics* 11 (2017), 453–556; arXiv:1512.06409.
- [23] F. Brown, C. Dupont, “Single-valued integration and superstring amplitudes in genus zero,” *Communications in Mathematical Physics* 382 (2021), 815–874; arXiv:1910.01107.
- [24] S. Caron-Huot, L. J. Dixon, J. M. Drummond, F. Dulat, J. Foster, Ö. Gürdoğan, M. von Hippel, A. J. McLeod, G. Papathanasiou, “The Steinmann cluster bootstrap for $\mathcal{N} = 4$ super Yang–Mills amplitudes,” *PoS CORFU2019* (2020), 003; arXiv:2005.06735.
- [25] G. Kings, J. Sprang, “Eisenstein-Kronecker classes, integrality of critical values of Hecke L -functions and p -adic interpolation,” *Annals of Mathematics* 202 (2025), 1–109; arXiv:1912.03657.
- [26] G. Kings, J. Sprang, “On the algebraicity and integrality of critical L -values of Hecke characters of imaginary quadratic fields,” survey, 8pp; arXiv:2511.05198.
- [27] V. Kufner, “On Deligne’s conjecture for algebraic Hecke characters of totally imaginary number fields,” preprint 2024, 49pp; arXiv:2406.06148.
- [28] K. Rémondère, “OP-NEWTON-DICKSON-SIGNED — Per-orbit twist character resolution via Gauss–Cox genus mechanism (1248/1248),” internal note, 2026-05-17.
- [29] R. Kotecký, D. Preiss, “Cluster expansion for abstract polymer models,” *Comm. Math. Phys.* 103 (1986), 491–498.
- [30] K. Rémondère, “OP-BEILINSON-Q-D — Beilinson regulator route to a $q(D)$ closed form,” internal note, 2026-05-17.
- [31] K. Rémondère, “OP-NEWTON-DICKSON-PROOF — Rigorous proof of the Newton–Dickson chain for $\mathbb{Q}$ -rat CM newforms,” internal note, 2026-05-17.
- [32] K. Rémondère, “Galois-Norm-Multiplicative Invariants of CM L -values and the Yang–Mills Period Problem,” Paper P4, 2026-05-17.
- [33] M. F. Paulos, J. Penedones, J. Toledo, B. C. van Rees, P. Vieira, “The S-matrix Bootstrap I: QFT in AdS,” *JHEP* 11 (2017), 133; arXiv:1607.06109.
- [34] M. F. Paulos, J. Penedones, J. Toledo, B. C. van Rees, P. Vieira, “The S-matrix Bootstrap II: Two Dimensional Amplitudes,” *JHEP* 11 (2017), 143; arXiv:1607.06110.
- [35] M. F. Paulos, J. Penedones, J. Toledo, B. C. van Rees, P. Vieira, “The S-matrix Bootstrap III: Higher Dimensional Amplitudes,” *JHEP* 12 (2019), 040; arXiv:1708.06765.
- [36] K. Rémondère, “Phase E — Extension ECI prédictions au-delà m_{0++} ,” internal note 2026-05-15 (memory snapshot `project_phase_e_eci_extension_motivic_weight`).
- [37] K. Rémondère, “Phase 8 ECI v15 état net,” internal note 2026-05-11 (memory snapshot `project_phase8_morn39_dayend_v12`).
- [38] J. Polchinski, *String Theory*, Cambridge University Press, 1998; vol. I, §6 on flux sectors and H^2 .
- [39] K. A. Ribet, “Galois representations attached to eigenforms with Nebentypus,” in *Modular Functions of One Variable V*, LNM 601, Springer, 1977, pp. 17–51.
- [40] A. J. Scholl, “Motives for modular forms,” *Invent. Math.* 100 (1990), 419–430.
- [41] M. Schütt, “CM newforms with rational coefficients,” *Ramanujan J.* 19 (2009), 187–205; arXiv:math/0511228.
- [42] G. Shimura, *Introduction to the arithmetic theory of automorphic functions*, Iwanami Shoten / Princeton University Press, 1971.
- [43] G. Shimura, “The special values of the zeta functions associated with cusp forms,” *Comm. Pure Appl. Math.* 29 (1976), 783–804.

- [44] K. Rémondrière, “W3+W4 Phase E EXTENSION FALSIFIED 2026-05-15,” memory snapshot `project_W3W4_phase_e_extension_falsified_2026-05-15`.
- [45] P. Borga, S. Cao, S. Shogren, S. Knaak, “Surface sums for lattice Yang-Mills in the large- N limit,” arXiv:2411.11676 (2025).
- [46] E. Costa, A. Elsenhans, J. Jahnelt, J. Voight, “Kuga-Satake constructions for sextic CM fields,” arXiv:2502.15052 (2025).
- [47] A. L. Guerrieri, A. Homrich, P. Vieira, “Multiparticle Flux Tube S-matrix Bootstrap,” *Phys. Rev. Lett.* 134 (2025), 041601; arXiv:2404.10812.
- [48] A. L. Guerrieri, A. Hebbar, B. C. van Rees, “Constraining Glueball Couplings,” *Phys. Rev. D* 112 (2025), 094023; arXiv:2312.00127.

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